

Involvement in Boundary-Spanning Activity: Mitigating the Relationship between Work Setting and Behavior

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ABSTRACT

While behavior is influenced to a large extent by the situation in which it occurs, it is argued that consequences of engaging in boundary-spanning activity are likely to diminish the impact of the internal work setting on boundary spanners' work behavior. To assess this premise, four hypotheses are proposed regarding the mitigating effects of boundary-spanning activity on the relationship between three work setting features and a set of behaviors. Findings indicate that this relationship is weaker among spanners than among other organizational members for one of the three work-setting variables. In addition, the mitigating effects increase as the amount of boundary activity increases. These results appear limited to in-person boundary activity, as opposed to boundary interactions over the phone, and the location of the boundary activity does not appear to be a relevant factor. Implications for managers and for scholars are discussed.

In order to perform well, organizations must cope with the demands and contingencies presented by an increasingly complex and uncertain environment. To respond to their environments effectively, organizations must engage in greater amounts of boundary activity. A growth in organizational boundary activity, for example, the set of organizational tasks that involve interaction between the organization and its environment, requires involvement of more and more of the organizations' members. Boundary spanners—individuals who interact with the environment across organizational boundaries—thus are playing an increasingly important role in organizational functioning. It is through their behavior that an organization adapts or fails to adapt to changes in the environment (Aldrich and Herker 1977).

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Boundary spanners are particularly critical to public sector organizations. In the public sector, clients or customers often receive services directly, with a greater proportion of personnel—the street-level bureaucrats (Lipsky 1980)—involved in actual service delivery. Interactions between clients and agency personnel are important influences on the level of client satisfaction, and they even impact citizens' perceptions of government as a whole (Kahn, Katz, and Gutek 1976). Since public organizations are accountable to the citizenry, more boundary activity is necessary to identify and respond to client dissatisfaction. Thus, many argue that public sector organizations must become more customer service oriented in order to be effective (e.g., Osborne and Gaebler 1992; Wagenheim and Reurink 1991).

Public sector organizations also require frequent interactions with other organizations. For example, they are involved frequently in various forms of interorganizational alliances (Miles and Snow 1986; Powell 1990), which are often necessary for effective service delivery (Lawless and Moore 1989). The personnel involved in these efforts engage in significant amounts of boundary interaction with each other (Gray 1985; Mandell 1984). Also, legislative oversight and other forms of regulation require boundary activity designed to address the concerns and demands of those monitoring the organization. As such, boundary spanning is a critical contingency affecting an organization's ability to mediate the effects of external political authority (Bozeman 1987).

In a number of ways, then, boundary activity is critical to the effectiveness of public sector organizations. Hence, it would be useful to have a better understanding of how this activity affects important organizational dynamics. Toward this end, this research explores the issue of how involvement in boundary-spanning activity impacts the influence of the organizational work setting on individuals' job behavior. While much job behavior can be viewed as a function (at least in part) of various situational characteristics, it is argued that involvement in boundary-spanning activity will mitigate the impact of the internal work-setting characteristics on behavior. This is because involvement in boundary spanning creates a set of dynamics that serves to reduce the influence of the internal work setting as a determinant of behavior.

The theoretical basis for this study is discussed in the next section. First, the concept of boundary-spanning activity is discussed, and the definition used in this study is articulated. Second, a rationale is provided for why organizational members'

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job behavior is likely to be influenced by the work setting in which it occurs. Third, an argument is made as to why involvement in boundary-spanning activity will weaken the influence of the internal work setting on job behavior.

THEORETICAL FOUNDATIONS

Definition of Boundary-Spanning Activity

Boundary-spanning activity entails the interactions between an organization and its environment. A number of specific boundary activities have been identified in the literature, including procurement of resources and disposal of outputs (Katz and Kahn 1966), exchange of information with the task environment (Leifer and Delbecq 1978), protection and buffering of the organization from external threat and pressure (Adams 1980), domain determination and interface (Jemison 1984), linking and coordination of activities between organizations (R.H. Miles 1980), external representation (Aldrich and Herker 1977), and relating of the organization to its larger community or social system (Katz and Kahn 1966). This list is not necessarily exhaustive, but it does reflect the diversity among the potential boundary-spanning activities.

Of course, individuals, not organizations, interact with the environment by engaging in boundary activities (R.H. Miles 1980; Organ 1971b). The set of individual tasks that comprises these activities is undoubtedly very large and diverse. Hence, specification of a defining characteristic of boundary activity would help identify boundary spanners for research purposes. Since boundary activity is defined at the organizational level as interaction with the environment, it is reasonable to utilize this criterion at the individual level also. Thus boundary activity is defined as any task in which an organizational member engages in job-related interaction with a person who is considered part of the environment, that is, who is not a member of the organization. In turn, organizational boundary spanners can be identified as those members who engage in any boundary activity (cf, Leifer and Delbecq 1978).¹

This definition of boundary activity is consistent with previous literature. Kahn et al. (1976, 180) discussed transactions between organizations and their environments in terms of "the smallest meaningful social transaction: a discrete episode involving two persons," one who acts as a member of the organization and the other who does not. Thompson (1962) analyzed boundary spanning in terms of interpersonal interactions between

¹This definition ignores differences in the amount or regularity of organizational members' boundary activity. One might argue that a person should not be considered a boundary spanner unless he or she engages in some baseline amount of boundary activity and/or unless there is some degree of regularity in the boundary activity. However, the intent here is to adopt a conceptualization of boundary spanners that is based on a well-defined and justifiable criterion and yet is as generic as possible given that constraint. Incorporating some qualifying level of activity into the definition provided here would undermine this effort. Furthermore, since the choice of any baseline level greater than zero would be rather arbitrary, the most defensible approach is simply to distinguish between those who do and those who do not engage in boundary activity.

member and nonmember, and Brown (1966) identified boundary activities as those in which people deal with outsiders. Organ (1971a and 1971b) and Jemison (1984) also viewed boundary spanning as involvement in interactions with others outside the organization.

This conceptualization puts into a single general category a broad array of members who engage in a diverse set of boundary tasks of varying importance to the organization.² The differences among various subcategories of boundary spanners are certainly not unimportant, and they deserve further exploration. However, the purpose of this research is to study boundary spanners in general, for example, to examine whether engaging in *any* type of boundary activity impacts the relationship between internal work setting and job behavior. Despite the many differences in function or significance, boundary spanners share in common at least one factor. They all interact with outsiders, and this interaction has important ramifications relevant to the role the internal work setting plays as an influence on work behavior. A discussion of these ramifications is provided below, after a brief discussion of the relationship between work setting and behavior.

Impact of the Work Setting on Behavior

A basic premise in the study of organizations and administration is that the behavior displayed by individuals in an organization is influenced greatly by the setting in which they work. An individual's work setting can be defined as the constellation of social and physical factors, internal to the organization but external to the person, that surround an individual on the job and that he or she experiences during involvement in the organization. These would include such factors as the formal structural and administrative systems (e.g., organization structure; reward system), the informal aspects of the social system (e.g., culture; management style; group dynamics), characteristics of the technological systems (e.g., work flow design; job design), and the nature of the physical setting (e.g., space configuration; interior design; cf. Porras and Robertson 1992).

Behavior is influenced by the internal work setting because organizational members' perceptions regarding what types of behavior are appropriate and/or feasible are determined to a considerable extent by work-setting characteristics. In other words, the internal work setting serves as a source of the behavioral expectations perceived by the organizational member, and individuals utilize this information in their decisions regarding how to act. Control of the appropriate stimuli and reinforcers

²This conceptualization also makes no distinction between public and private sector boundary spanners. Literature specifically oriented toward an analysis of boundary spanning in public sector contexts is very limited, as indicated by the fact that electronic searches of four databases of periodical literature yielded very few references to articles of this nature. While public sector boundary spanners certainly have been the focus of considerable research, attempts rarely have been made to ascertain the particular implications of the public sector nature of the organizations involved. Thus it is not yet possible to derive any empirically based generalizations about potential differences associated with boundary spanning in the public as opposed to the private sector. Furthermore, given recent analyses that suggest the lines between the two sectors are quite blurred at best (Bozeman 1987; Perry and Rainey 1988), it is not clear that any such differences should even be expected.

thus can affect significantly the behavior of individuals at work (Hackman 1981). Consequently, the design of organizational systems that will generate predictable and constructive participant behavior is of critical importance to those responsible for managing these participants.

The focus here is on the types of behavior individuals engage in, rather than their level of effort (cf, Naylor, Pritchard, and Ilgen 1980). The former may be more important for organizational success than the latter (Katerberg and Blau 1983). Specific types of individual behavior recently have been identified in the literature. Organizational citizenship behaviors (Graham 1989; Organ 1988; Smith, Organ, and Near 1983) comprise one set. Another set, defined by Porras and Hoffer (1986), consists of behavior patterns that change as a result of successful planned organizational efforts. Behaviors such as these often go beyond role prescriptions (Katz 1964), and they may be important factors shaping organizational performance (McAllister 1988). For example, Hoffer (1986) found strong relationships between the frequency of the behaviors outlined by Porras and Hoffer (1986) and various financial measures of organizational performance.

A better understanding of the impact of the internal work setting on behavior requires consideration of how specific organizational factors promote or elicit key types of behavior. Previous research (Robertson 1994) examined the relationships between three work setting features and a number of behavior categories identified by Porras and Hoffer (1986).³ In particular, that study assessed whether individuals engaged in the behaviors with greater frequency when their managers more frequently engaged in a set of managerial behaviors also identified by Porras and Hoffer (1986); their job designs were more "motivating" (Hackman and Lawler 1971); and/or their work goals were specified more effectively.

The present analysis extends this prior research by examining whether the relationship between work setting—operationalized in terms of the above three specific work setting features—and behavior is affected by the organizational member's involvement in boundary-spanning activity. It is predicted that involvement in boundary activity will mitigate this relationship, since dynamics typically associated with boundary spanning are likely to reduce the importance of the internal work setting as an influence on work behavior. The nature of these dynamics, as suggested by an integration of previous theory and research on boundary spanners, is discussed next.

³These behaviors will be described in greater detail.

Behavioral Implications of Boundary Spanning

Public-sector organizations constitute a complex arena in which boundary spanners must operate. This derives in part from the fact that a number of different mechanisms are utilized to control the activities of these organizations in their efforts to execute and implement governmental policy decisions. In particular, these organizations and their members are simultaneously subject to the rule of law, political control, and self-control through professionalism and expertise (Mainzer 1973). The first two of these reflect forces, external to an organization, that influence and constrain its activities. For example, Mosher (1978) points to trends of increasing citizen and politician influence on decisions that previously were left to the professionals inside the bureaucracy. Even enhanced professionalism is not simply an internal determinant of action, since the objectives and values of the various professions represented in an organization may not be completely consistent with those of the organization itself (Mosher 1982). Furthermore, the boundaries of many professions are enlarging to incorporate new elements and techniques, and many professionals are working more frequently with representatives of other professions on a problem-oriented and interdisciplinary basis (Mosher 1978).

As an organization strives to effectively carry out its activities, therefore, it must take into account a variety of pressures that emanate from its external environment. Boundary spanners, as the representatives of the organization in its interactions with the environment, experience most directly the specific requests, demands, and constraints of the particular outsiders with whom they interact. Two important features of boundary spanners' work experience derive from their interactions with outsiders, both of which are likely to reduce the impact of the internal work setting on boundary spanners' work behavior.

First, boundary spanners have role senders located externally as well as in their own organization (Keller and Holland 1975). In addition to representatives from other organizations with whom they must interact to fulfill their role requirements, these can include citizens and members of other interest or advocacy groups (Mainzer 1973), as well as other members of the boundary spanners' profession. Thus, boundary spanners must attend to divergent informational and social stimuli from various sources inside *and* outside their organizations (Dubinsky, Hartley, and Yammarino 1985), and they often interact with people from groups whose norms differ from one another (Caldwell and

O'Reilly 1982). As a result, boundary spanners often find that they are the focus of inconsistent expectations from different groups (Kahn, Wolfe, Quinn, and Snoek 1964), and they are expected to be accountable to potentially incompatible sources of influence, for example, agency preferences and client claims (Lipsky 1980). Behavior that is most instrumental in these external interactions may represent violations of the norms and values of the internal constituency (Organ 1971a).

By attending to conflicting social cues, behavioral norms, and role expectations, boundary spanners can become more uncertain about how to perform their job responsibilities. For example, Dubinsky et al. (1985) found that role ambiguity is related positively to the extent to which boundary spanners are sensitive to the various cues present in their social situations. This ambiguity is likely to reduce the extent to which boundary spanners' behavior is consistent with cues derived from their internal work settings. Furthermore, while interaction with individuals in one's own organization reinforces organizational norms, interaction with outsiders can lead to the development of nonorganizational norms regarding behavior. This also suggests that a weaker link would be found between the internal work setting and behavior for boundary spanners than for nonspanners.

Second, boundary spanners are often psychologically and/or physically closer to the outsiders with whom they interact than to internal constituents (Adams 1976). Hence, internal constituents can find it difficult to monitor boundary spanners' behavior and evaluate their performance (Lipsky 1980). As a result, internal expectations regarding appropriate behavior may be less important to boundary spanners than to other members (Organ 1971a). Boundary spanners may be less familiar with these expectations, as there may be fewer opportunities for them to be communicated or to become internalized. Or they may not be fully accepted, since greater distance can weaken boundary spanners' organizational bonds or the importance they place on organizational membership (Adams 1976).

These factors are likely to reduce the extent to which boundary spanners behave in a manner consistent with expectations generated from the internal work setting. For example, conformity to internal constituents' behavioral prescripts decreases when boundary spanners perceive that the constituents' ability to monitor their behavior decreases (Organ 1971a). If the boundary spanner assumes that expectations regarding behavior will not be a criterion by which he or she is evaluated, conformity to these expectations will be low. Alternatively, if the

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boundary spanner has strong bonds to the organization and continued membership is desirable, it is likely that he or she will act in ways perceived as appropriate, both out of loyalty and out of a strong desire to appear loyal (Adams 1976).

In sum, because boundary spanners interact with the organization's environment, their behavior is cued by both organizational and environmental factors. They often are subjected to greater pressure from external sources than are persons occupying internal, buffered roles (Miles 1976), and their efforts to respond to external expectations and norms should mitigate the impact of the internal work setting on their behavior. The organization's ability to communicate expectations and to monitor and control behavior is also frequently less than it is for other members. This should reduce the likelihood that boundary spanners will engage in behavior that conforms to internally generated signals.

Hypotheses

Four hypotheses are explored in this study. The first is derived directly from the basic premise, outlined above, that involvement in boundary-spanning activity mitigates the strength of the relationship between organizational members' internal work setting and their work behavior.

Hypothesis 1: The relationship between work setting and behavior will be weaker among boundary spanners than among other organizational members.

In addition to the above comparison, boundary spanners can be examined in greater detail to evaluate whether characteristics of their boundary activity influence the mitigating effects of this activity. Three additional hypotheses are based on the idea that interaction with external individuals can vary on a number of dimensions that might affect the degree to which this boundary activity reduces the impact of the work setting on behavior. The most obvious characteristic is the amount of time spent in such activity. All else equal, the consequences of engaging in boundary activity are likely to be more potent as the amount of boundary activity increases.

Hypothesis 2: The strength of the relationship between work setting and behavior will be associated negatively with the amount of boundary-spanning activity.

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Boundary interactions with outsiders can take place either in person or by telephone. It is likely that the various factors described above as impacting boundary spanners will result more from in-person interactions than from phone interactions. The relatively impersonal nature of telephone interactions should make it easier for boundary spanners to discount the importance of outsiders' behavioral expectations, or at least it will make it harder for the outsiders to monitor their behavior. Likewise, interacting primarily over the phone will limit the extent to which boundary spanners feel more distant from and less tied to their own organizations. Finally, since most telephone interactions are likely to occur in their own work areas (i.e., the work areas designated for them individually or allocated to the work unit of which they are members), the relatively greater exposure to this work setting and the internal behavioral expectations will remain high, and thus behavior should be more consistent with these expectations than those generated by outsiders. In contrast, interacting with outsiders in person should serve to reduce the importance of the organizational work setting as an influence on the boundary spanners' behavior.

Hypothesis 3: The strength of the relationship between work setting and behavior will have a stronger negative association with the amount of boundary-spanning activity occurring in person than with the amount of boundary-spanning activity occurring by telephone.

The location of boundary interactions may influence the degree to which boundary activity impacts the relationship between the organizational work setting and work behavior. As indicated above, one difference between the two categories of boundary interactions is that in-person interactions are more likely than telephone interactions to occur in places other than the individual's own work area. The more individuals are away from their own work settings, the less likely it is that those settings will influence their behavior. Furthermore, the more that boundary interactions take place in outsiders' organizations, the greater is the possibility that boundary spanners will feel more distant from the organization and that organizational bonds will be weakened. Such factors will serve to increase the salience of the outsiders' behavioral expectations and reduce the impact of the internal work setting on behavior.

Hypothesis 4: The strength of the relationship between work setting and behavior will have the strongest negative association with the amount of boundary-

⁴Missing data typically resulted from the failure of the respondent to answer one or two questions on the survey. In the analysis (using SPSS-X), the "listwise deletion of missing data" option was chosen, which drops from the analysis any case that is missing data on any variable. While the downside to this approach is that it drops a larger number of responses than would other options (e.g., replacing missing data with the mean for that variable), it is the more conservative approach. It does not seem likely that the amount of missing data reflects a problem with the instrument. Any problems with the wording of questionnaire items were identified in the pretest phase and corrected. Also, there were no patterns across respondents regarding which particular survey items were left unanswered. Thus, failures to respond seem to have been idiosyncratic to the individual respondents. In any case, t-tests indicated there were no differences on the research variables between the responses of those included in the analysis and those excluded from the analysis.

⁵This exclusion was deemed acceptable for three reasons. First, the theoretical mechanisms thought to underlie the relationships under investigation in this study are more likely to occur as a result of vocal interactions than nonvocal interactions. In other words, the former are more likely than the latter to generate external role pressures, greater distance from the organization, and the associated consequences of these two factors. Second, the jobs held by respondents in this sample were such that most of the boundary activity took the form of vocal interaction with outsiders. Only a small percentage of the interaction with outsiders consisted of interaction through nonvocal media. Third, this measure is equivalent to operationalizations that have been used in previous research. For example, Tushman (1977; Katz and Tushman 1983; Tushman and Scanlan 1981) measured boundary spanning in terms of the amount of work-related oral communication with external professionals, ruling out written and social communication. Dollinger (1984) assessed the number of contacts boundary spanners

(note 5 continues)

spanning activity occurring in outsiders' organizations; the weakest negative association with the amount of boundary-spanning activity occurring in boundary spanners' own work areas; and a medium association with the amount of boundary-spanning activity occurring in other locations.

METHODS

Sample and Data Collection

The sample for the research consisted of 105 nonsupervisory employees of two public sector organizations. The first was the supply department of a military base, in which both civilian and military personnel were distributed across sixteen work centers. The second organization was a regional office of a national regulatory agency, in which employees in three broad job classifications comprised twenty-six work units. The survey instrument used to collect data was pretested by having ten employees from each organization fill out the survey and provide feedback on the overall design of the instrument and on any items that were poorly worded or otherwise confusing. Changes to the instrument were made based on this feedback, and these individuals were excluded from the actual data collection.

The final version of the survey instrument was distributed to seventy-five employees in the military base supply department, and forty-two were returned. In the regulatory agency office, a meeting was held in which all employees were given the instrument, yielding ninety completed surveys. Of the 132 surveys received, twenty-seven were excluded from the analysis because of missing data.⁴ To determine whether there were any differences in the responses of the employees from the two organizations, t-tests were used to compare the means of the two groups on the work-setting variables and the work behavior variable. None of the differences between the means were significant, and thus the responses from the two organizations legitimately could be combined into a single sample.

Measures

Boundary-spanning activity. Boundary-spanning activity is defined as any task in which an organizational member engages in job-related interaction with a person who is not a member of the organization. For this study, the focus was limited to vocal interaction. Interaction through nonvocal means such as letters, memos, or electronic mail was not included.⁵ Respondents were asked: In carrying out your job responsibilities, do you ever talk,

³(continued)

had and the percentage of time they spent in the average week with each of nine categories of environmental constituencies. Likewise, Ito and Peterson (1986) measured the frequency of and time spent in interactions with nonmembers.

It might be argued, however, that this operational definition is too restrictive in terms of the modes of communication included, for instance, that activities other than vocal interaction with outsiders can be construed as boundary activity. Some would include written communication with outsiders as being a boundary activity (e.g., Lyonski 1985). Furthermore, some consider boundary spanning to include any form of information acquisition, such that boundary activity could include simply acquiring information from newspapers, magazines, journals (e.g., Keller, Szilagyi, and Holland 1976). While this argument has some legitimacy, it is also problematic in that there are no clear guidelines for delineating between information acquisition which is boundary activity and that which is not. At the extreme, anyone who acquires some organizationally relevant information from whatever source becomes a boundary spanner, with the result that nearly every organizational member could be identified as a boundary spanner. Hence, it would not provide a criterion by which to differentiate boundary spanners from nonspanners. In contrast, limiting the operational definition to vocal interaction with outsiders does provide a clear cut, theoretically justifiable criterion by which to make this distinction.

⁴It might be argued that the operationalization of boundary spanners used here is too broad in terms of the frequency of boundary interactions—that engaging in any boundary activity, however infrequent, is sufficient to label someone a boundary spanner. This argument could be based on the fact that the definition results in nearly three-fourths of the sample being identified as boundary spanners. As indicated earlier (note 1), a narrower definition might require that an individual engage in some baseline level of boundary activity (e.g., amount or regularity) before being considered a boundary spanner. Again, however, any
(note 6 continues)

either in person or over the phone, with people who are not employees of your organization? Boundary spanners were distinguished from nonspanners by their indication that they did engage in job-related interaction with outsiders. Of the 105 respondents, seventy-seven identified themselves as boundary spanners, and twenty-eight identified themselves as nonspanners.⁶

For the boundary spanners, four aspects of their boundary activity were assessed. The first two, the amount of in-person boundary activity and the amount of over-the-phone boundary activity, were computed as the average number of work hours spent per week interacting in person and on the phone, respectively, with individuals not members of the respondent's organization, divided by the average number of hours spent per week at work. Third, the total amount of boundary activity was computed as the sum of the in-person and over-the-phone measures. Finally, the location of the in-person boundary interactions was assessed by calculating the number of these interactions (defined in terms of the percentage of hours spent per week at work) occurring in each of three locations, namely the respondent's own work area, somewhere in the respondent's organization other than his or her own work area, and in the work area of the person with whom the respondent interacted.

Work setting. While the hypotheses include *work setting* as one of the constructs to be studied, a work setting clearly comprises a number of specific elements that potentially can influence the behavior of organizational members. Therefore, to measure work setting, it is necessary to operationalize it in terms of specific features that shape the context in which individuals behave at work. As indicated above, the three features included in this research are the behavior patterns of the respondents' direct managers, the design of their jobs, and the goals of their work groups. These features were selected because they met two important criteria. First, there is a priori justification, based on existing theory and research, for expecting a relationship between these work-setting features and the behaviors examined in this research.⁷ Second, these features each have been a frequent target of organizational change efforts designed to improve individual behavior or performance. The measures for each of these were derived from responses to the survey questionnaire. The specific survey items comprising the indices described below are provided in the appendix.

Porras and Hoffer (1986) identified a set of five key managerial behavior categories: promoting information flow, functioning strategically, developing others, leading by vision, and generating participation. Hoffer (1986) developed questionnaire

⁶(continued)

other choice of a baseline criterion would be essentially arbitrary and would be inconsistent with the intent of this research. Thus rather than concluding that the operational definition is too broad, the high proportion of boundary spanners in this sample can be viewed simply as an indication that boundary spanners are in fact quite prevalent in many organizations.

⁷It is beyond the scope of this article to provide the detailed rationales for why these three work-setting variables are expected to be related to employee behavior. However, the interested reader can find these explanations in Robertson (1994).

⁸In the original research (Robertson 1994), scales reflecting separate dimensions of the three work-setting variables and the work-behavior variable were created to enable more fine-grained analyses of the relationships between work setting and behavior. However, analyses utilizing these scales are beyond the scope of this article. Therefore, the indices for all these variables are computed as the means of the relevant scale scores.

⁹In contrast to the other two work-setting variables, there was no single source for the questionnaire items used to assess goal effectiveness. Instead, through a review of literature on both goal setting (e.g., Latham and Locke 1979; Latham and Yukl 1975) and management by objectives (e.g., Albrecht 1978; Lee 1981; Odiorne 1979), a number of goal characteristics were identified that are believed to be important for goals to affect employee behavior in desired ways. Twenty-one original questionnaire items were developed with which to assess the nine dimensions identified by this review.

¹⁰The remainder of the items did not load strongly on any of the four factors derived from the factor analysis. Therefore, including them in these scales would have reduced scale reliabilities below an acceptable level. Furthermore, creating a separate scale or scales comprised of these items could not be justified by either the factor analysis results or a test (note 10 continues on p. 86)

items to assess these five categories. The first three categories were included in the present research. However, scales assessing the other two categories demonstrated insufficient reliability coefficients. A subset of the items from these two scales was combined (based on factor analysis results) to form a fourth scale, labeled providing guidance. Twenty items measured these four managerial behavior categories. The managerial behavior index is the mean of the four scale scores.⁸ This index has a reliability coefficient of .95.

Respondents' job design characteristics were measured using ten items from the job diagnostic survey developed by Hackman and Oldham (1975). Two questions measured each of five dimensions of job design: autonomy, skill variety, task identity, task significance, and feedback from the job. The job design index is the mean of the five scale scores, with a reliability coefficient of .63.

Twenty-one questions were developed to assess nine qualities of work-group goals suggested in the literature to be characteristic of effective goals.⁹ These qualities were familiarity, specificity, difficulty, congruency, importance, impact on job behavior, impact on amount of competition with others in the organization, motivational quality, and process of determination. A factor analysis identified four underlying dimensions—behavioral specificity, outcome specificity, motivational quality, and congruency—which comprised a total of fourteen questionnaire items.¹⁰ The goal effectiveness index is the mean of these four scale scores. The reliability coefficient of this index is .85.

Work behavior. The work behaviors measured are a subset of those identified by Porras and Hoffer (1986), which are described in exhibit 1. Hoffer (1986) developed questionnaire items to measure the frequency of these nine behavior categories. However, in her analysis of the relationship between these behaviors and organizational performance, she combined two categories—solving problems effectively and inquiring—into a single scale, labeled solving problems strategically, and she split the respecting/supporting others category into two separate scales. These changes were utilized in the present research, although the respecting scale was excluded from the instrument because of high response bias found in the pretest phase. In addition, the collaborating and experimenting scales subsequently were dropped from the analyses because of inadequate scale reliabilities. Forty-four questionnaire items measured the frequency of the remaining six behavior categories. A work behavior index was computed as the mean of the six scale scores, with a reliability coefficient of .85.

Exhibit 1
Categories of Behavioral Changes Resulting from
Successful Organizational Change Efforts

BEHAVIOR CATEGORY	DESCRIPTION
Communicating openly	Behaviors promoting or reflecting the direct giving and receiving of information relevant to getting the job done
Collaborating	Behaviors promoting or reflecting the involvement of relevant persons in the processes of identifying and solving problems
Taking responsibility	Behaviors reflecting acceptance of responsibility and taking initiative in carrying out organizational tasks
Maintaining a shared vision	Behaviors reflecting a clear formulation of, understanding of, and commitment to organizational philosophy, values, and purposes and a commitment to high standards
Solving problems effectively	Behaviors reflecting a problem-solving orientation to difficult organizational issues
Respecting/supporting	Behaviors demonstrating respect and support for others as worthwhile individuals
Processing/facilitating interactions	Behaviors reflecting attention to and use of human process issues in one-on-one, group, and intergroup situations
Inquiring	Behaviors reflecting a probing, inquiring, diagnostic orientation to the organization and its environment
Experimenting	Behaviors promoting or reflecting an openness to trying new things

Adapted from Porras and Hoffer 1986.

Relationship strength. To assess the strength of the relationship between work setting and behavior, work behavior was regressed on managerial behavior, job design, and goal effectiveness. For each respondent, the absolute value of the residual from this regression is a measure of how well behavior is predicted by the three work-setting variables, and thus provides an index for relationship strength. The absolute value is utilized, since the direction of the error in prediction is unimportant. The smaller the absolute value of the residual, the stronger the relationship between work setting and behavior.

Analyses

Two types of multiple regression analyses were used to assess hypothesis 1, that the relationship between work setting and behavior would be weaker among boundary spanners. In the first, work behavior was regressed on managerial behavior, job design, and goal effectiveness, with separate regressions for boundary spanners and for nonspanners. This allows a comparison of these two groups in terms of the overall strength of the relationship between the three work-setting variables and behavior.¹¹ For the second regression, a dummy variable was utilized to represent whether or not the respondent was a boundary spanner (coded 1 for spanners and 0 for nonspanners). Three interaction terms were created, as the products of the dummy variable and each work setting variable. Work behavior was then regressed on the dummy variable, the three work-setting variables, and the three interaction terms. The regression coefficients for the interaction terms, which reflect the differences between spanners and nonspanners regarding the relationships between the work-setting features and behavior, can be examined to evaluate support for the hypothesis.¹² A significant negative coefficient means that the relationship is weaker among boundary spanners.

To assess the three hypotheses regarding the impact of boundary activity characteristics on the relationship between work setting and behavior, the boundary activity measures were correlated with the relationship strength index. Since a larger regression residual indicates a weaker relationship between work setting and behavior, *positive* correlations between the boundary activity measures and the relationship strength index would support the hypotheses regarding a negative association between boundary-spanning activity and the strength of the relationship. It would indicate that boundary activity reduces the degree to which behavior can be predicted from the work-setting variables. Differences among the correlations provide evidence regarding the relative impact of in-person versus phone interactions and of the various locations of the boundary interactions.

¹¹(continued)

of face validity. Given the original intent to develop a set of scales for each work setting variable (see note 8), there was no rationale for including these items in the analysis, and thus they were omitted.

¹¹The multiple R^2 s and the regression coefficients from the two regressions can be compared only informally to evaluate the extent to which involvement in boundary spanning influences the strength of the relationship between the work-setting variables and behavior. There are no tests of significance to formally assess differences in the R^2 s and in the coefficients from the two regressions.

¹²This regression is equivalent to an analysis of covariance, where the work-setting features are the covariates. After controlling for any differences between boundary spanners and nonspanners (through the use of the dummy variable) and for the main effects of the work-setting variables, it examines whether there is an interaction between boundary spanning and the work-setting features.

RESULTS

Results for the two regressions in the first analysis are provided in exhibit 2. Looking first at the multiple R^2 s from the two regressions, the three work-setting variables combined predict more of the variance in work behavior among boundary spanners ($R^2 = .26$, $p < .001$) than among nonspanners ($R^2 = .16$, n.s.). Thus these results run counter to hypothesis 1. Of the three work-setting variables, the coefficient for goal effectiveness is significantly positive in the regression for boundary spanners, which again is in the opposite direction as hypothesized. The other two variables did not yield significant coefficients in either regression.¹³

Exhibit 2
Results from Multiple Regressions on Work Behavior

	FIRST ANALYSIS		SECOND ANALYSIS
	Boundary Spanners (n = 77)	Non-spanners (n = 28)	(n = 105)
Multiple R^2	.26	.16	.23
significance of F	(.00)	(.23)	(.00)
Managerial behavior	-.06 ^a (.62) ^b	.37 (.07)	.42 (.02)
Job design	-.10 (.43)	.09 (.68)	.13 (.61)
Goal effectiveness	.57 (.00)	.00 (.99)	.00 (.99)
Spanners vs. nonspanners			-.04 (.95)
MB interaction			-.69 (.03)
JD interaction			-.47 (.44)
GE interaction			1.16 (.05)

^aThe standardized beta coefficient from the regression.
^bThe significance level of the t-test for that coefficient.

¹³It is important to note that the sample size is only twenty-eight in the regression for the nonspanners. While this sample size may be adequate to have reasonable confidence in the results, it approaches a minimally acceptable level, and thus the results should be interpreted with caution.

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The second regression, results for which are also provided in exhibit 2, is highly significant overall ($R^2 = .23$, $p < .001$). It is worth noting first that the dummy variable coefficient is not significant, indicating that boundary spanners and nonspanners do not differ in terms of the frequency of these behaviors. Of the three main effects, only the managerial behavior coefficient is significant ($p < .05$), suggesting that this variable demonstrates the strongest positive relationship to these behaviors regardless of involvement in boundary activity. Regarding support for hypothesis 1, the evidence is mixed. The managerial behavior interaction term is significantly negative ($p < .05$) and thus in the predicted direction. However, the goal effectiveness interaction term is significant ($p = .05$) and positive, indicating a *stronger* relationship among boundary spanners than among nonspanners. The job design interaction term is not significant.

The analyses for the remaining three hypotheses were limited to the boundary spanners in the sample, since the focus was on the nature of their boundary activity. The correlations between relationship strength and the three measures of amount of boundary activity (total, in person, and over the phone¹⁴) provide support for hypotheses 2 and 3. First, there is a significant correlation between relationship strength and the total amount of boundary activity in which an individual engages ($r = .27$, $p < .01$), indicating that the work-setting variables predict behavior less accurately as the total amount of boundary activity increases. Furthermore, it is apparent that these effects are tied primarily to the amount of in-person interaction, since this correlation is significant ($r = .27$, $p < .01$) while the correlation involving phone interactions ($r = .13$) is not.

In contrast, the correlations between relationship strength and the amount of in-person boundary activity taking place in each of the three locations fail to support hypothesis 4. None of the correlations are significant and they are all relatively equal—.12 for the amount of boundary activity in the respondent's own work area, .12 for the amount elsewhere in the boundary spanner's own organization, and .10 for the amount in the other person's work area. These results suggest that the location of the boundary interactions has very little impact on the relationship between work setting and behavior.

DISCUSSION

The basic premise of this study, that involvement in boundary-spanning activity mitigates the impact of the internal work setting on work behavior, received mixed support. The

¹⁴These respondents reported that they spent an average of 35 percent of their work week engaged in boundary activity, 13 percent in in-person boundary interactions and the remaining 22 percent on the phone.

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findings suggest that the effects of boundary activity are not uniform in two important ways. First, as predicted, the effects depend on the amount and type of boundary activity. As individuals engage in more boundary activity, especially in-person interactions, the mitigating effects of this activity increase. Second, the effects vary across different features of the work setting; this finding was not anticipated. While some features (managerial behavior) demonstrate weaker relationships with behavior among boundary spanners, others (goal effectiveness) demonstrate stronger relationships, or there are no differences at all (job design). Although this study measured only three specific work setting variables, it is certainly possible, maybe even likely, that the pattern of findings among these three variables can be generalized to a much broader range of work-setting features. In other words, among the full range of work-setting features, some may influence the behavior of boundary spanners less than that of nonspanners while other features may have a stronger impact on boundary-spanner behavior.

It is important to consider possible explanations for the pattern of results found in this study. First, there are a number of reasons a manager's behavior would not be as important an influence on boundary spanners as on nonspanners. Spanners frequently spend much of their work time engaged in interactions away from the presence of their managers. Therefore, there is often less exposure to the influence of the manager, and/or this influence is less salient to spanners as they decide how to behave on the job. Also, much public-sector service delivery behavior is hard to monitor or assess, and boundary spanners may be the only source of information regarding their own performance effectiveness (R.H. Miles 1980). Managers may thus find it hard to exert much direct control or influence. Furthermore, since street-level bureaucrats expect to make important discretionary decisions themselves, their managers' efforts to dictate service norms may be regarded as illegitimate (Lipsky 1980). Lipsky points out that these boundary spanners frequently possess various resources that enable them to resist managerial pressures regarding job performance.

As a result, the behavior of boundary spanners appears to be influenced more by the nature of their work goals than by their managers' behavior. In fact, goals are related *more* strongly to boundary spanner behavior than to the behavior of nonspanners. One possible explanation may be that boundary spanners find it difficult or undesirable to conform to outsiders' expectations. In public sector organizations, for example, there is often an extensive set of rules and regulations regarding performance and/or

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strong norms regarding such behavioral concerns as equal or equitable treatment of clients. Thus the extent to which spanners' behavior is shaped by external factors, including the expectations of outsiders, may be limited. As a result, important internal factors *other than* their managers may help them determine the most appropriate type of behavior.

One such factor would be their work goals. In the midst of possible ambiguity regarding appropriate job behavior and in the absence of reliance on signals from their managers, boundary spanners may behave in ways that they perceive will most effectively help them achieve their goals. While a manager may not always be present to monitor and influence behavior, goals can serve as a continuous source of direction regarding appropriate job behavior. A number of different types of goals serve to shape boundary spanner behavior in public-sector organizations. These range from specific, quantitative, productivity-related goals, to broader goals concerning the quality of treatment or interaction with organizational outsiders, to very abstract goals pertaining to, for example, a public service orientation (cf, Perry and Wise 1990).

Unlike the first two variables, job design is not related to the behavior of either boundary spanners or nonspanners. One potential explanation is that job design affects attitudinal or motivational responses more than it impacts behavioral responses (Pierce and Dunham 1976; Staw 1984). Alternatively, it could be that job design impacts different behaviors than those assessed here (e.g., Luthans, Kemmerer, Paul, and Taylor 1987). Third, the lack of a relationship between job complexity and behavior among either group may reflect the fact that relevant moderating variables—namely level of growth need strength, knowledge and skill, and context satisfaction (Hackman and Oldham 1975)—were not included in this study. Research (Fried and Ferris 1987) indicates that growth need strength does moderate the relationship between job characteristics and job performance.

Our findings support the prediction that among boundary spanners the relationship between their organizational work setting and their work behavior is weaker when they engage in a greater amount of boundary activity. This is true particularly when the boundary activity entails in-person interactions with outsiders. Boundary activity occurring by telephone does not have a similar mitigating impact on the relationship between work setting and behavior. Likewise, there is no evidence that the location of the in-person boundary interactions is related to

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the extent to which behavior is associated with the nature of the internal work setting.

The latter finding suggests that the mitigating effects of involvement in boundary-spanning activity can occur regardless of where the boundary interactions take place. One interpretation is that it is the existence of the interaction itself, more than the exposure to an external environment (and consequent freedom from the internal work setting), that serves to reduce the impact of the internal work setting on behavior. Consistent with this notion, Adams (1980) points out that boundary interactions constitute a boundary system, such that the interpersonal interactions between boundary spanners from two different organizations develop a social structure with its own attributes, behavioral norms, and so forth. Boundary spanners' work behavior is influenced by the components of these systems and the relationships between these components, only one category of which are the attributes of the internal work setting. Since the characteristics of these boundary systems would develop regardless of the location of the interactions, the critical dynamics are probably associated with the mere existence of these systems, rather than where the interactions take place. Furthermore, it is not surprising that the impact of the work setting appears to be weaker when the amount of boundary activity is greater, as this would strengthen the development of an influential boundary system.

Implications

The implications of this study for managing public sector boundary spanners are rather straightforward. First, a critical task for public managers is to establish and clarify key group and organizational goals that effectively guide employee behavior. A frequent assertion in the literature on public organizations is that they pursue multiple, intangible, conflicting goals (Rainey 1991). Such goal diffuseness inevitably affects organizational members, as priorities remain unclear and guidelines for behavior are missing. The present findings suggest that effective articulation of goals may have a significant impact on the behavior patterns of boundary spanners, who are an increasingly large and critical segment of many organizations. As suggested above, goals need not simply be quantitative. Individuals in public sector organizations often are motivated by the qualitative, value-oriented goals of their work units, since many public programs have high intrinsic value (Stahl 1983). Furthermore, development of a customer service orientation as a fundamental organizing and operating strategy can serve to guide the actions of employees in desired directions (Wagenheim and Reurink 1991).

According to our findings, the use of goals to motivate and guide behavior may be a more effective strategy than the use of more directive approaches. R.H. Miles (1980, 320) suggests that managers should "pursue a more indirect [approach] of developing a strong interpersonal bond with the boundary spanner as well as a strong bond of loyalty and commitment between the boundary spanner and the organization and objectives he represents." This perspective is consistent with literature on leadership (e.g., Bennis 1989; Bradford and Cohen 1984; Burns 1978) that points out the importance of clarifying a vision or overarching goal, gaining commitment to the goal, and empowering individuals for goal achievement. Since managerial behavior has less direct influence on the behavior of boundary spanners, managers can increase their leverage by insuring that spanners are committed to and rewarded for the achievement of appropriate goals. If managers do not utilize this source of influence, they risk the consequence that boundary spanners will conform to externally rather than internally generated performance expectations. To the extent these result in undesirable behavior, organizational performance may suffer.

This study also has implications for theory and research. First, it contributes to a better understanding of how specific behaviors of different organizational members may be influenced by work-setting features. Yet more research of this nature is necessary in order to develop a sound theoretical framework delineating the impact of organizational work settings on employee behavior. A related question has to do with what types of behaviors are desirable for different types of employees. Porras and Hoffer (1986) attempted to identify a set of effective behaviors that would be relevant for any employee in any type of organization. Yet further research would be valuable for ascertaining the extent to which this assumption is accurate. In particular, it would be useful to determine whether these behaviors are in fact important to the effective job performance of public sector organizational boundary spanners. Kahn et al. (1976) indicate that clients evaluate their interactions with agency personnel based on four criteria: the extent to which it is difficult to find officials who can handle their problems; the effort exerted by agency personnel to help them; the extent to which they have a choice about agency activity or simply have to take what the agency personnel decide; and whether there is someone at a higher level in the office to whom they can appeal if their concerns are not adequately addressed. Similarly, Lipsky (1980) suggests that clients seek three values from service bureaucracies: service or benefits appropriate to their situations or needs; equity tempered by flexibility in the distribution of public benefits; and

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respect as citizens receiving their due from government. Future research should assess the extent to which greater use of the behaviors assessed in this study enables boundary spanners to enhance the quality of their interactions with clients and other outsiders.

The present research contributes to the literature on organizational boundary spanners by focusing on boundary spanning as a moderator variable rather than as either an independent or a dependent variable. Most research on boundary spanning has examined its consequences, either for the individuals engaged in boundary activity or for the groups or organizations of which they are a part. Additional research has attempted to identify antecedent conditions that influence the creation of boundary roles, the characteristics of these roles, and the behavior of boundary spanners. As a result, our understanding of boundary role characteristics and their antecedents and consequences is relatively well developed. In contrast, our knowledge of how boundary spanning moderates relationships between other types of important organizational variables is more limited.

Since the data do not completely support the hypotheses generated regarding the moderating effects of boundary-spanning activity, more research on this topic is warranted. Additional work-setting variables should be examined to explore more fully the variations in the impact of boundary spanning on the relationship between work setting and behavior. Other characteristics of boundary-spanning activity that contribute to its moderating effects also could be explored. Examples might include the number of outsiders with whom the boundary spanner interacts or the frequency and duration of these interactions. Research also is needed to assess the reasons involvement in boundary spanning affects the relationship between work-setting features and behavior. While a variety of role characteristics and their consequences were described earlier, the present study did not directly assess these factors. Thus future research specifically should examine the extent to which these dynamics are responsible for the moderating effects found in this study.

Finally, future research should explore these relationships among a variety of different types of boundary spanners. Organizational boundary spanners hold many different types of jobs and are responsible for a wide variety of organizational functions. Therefore, conducting research on boundary spanners in other types of organizations with a broader range of jobs would help clarify the extent to which patterns found in this study can be generalized, or whether the nature of the work impacts the

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degree to which boundary activity moderates the relationship between work setting and behavior. Also, research could explore whether there are any differences between public and private sector organizational boundary spanners that are relevant to the issues addressed in this research and those suggested above for future research.

To conclude, this research has important implications both for public managers who wish to manage effectively the behavior and performance of their boundary spanners and for scholars who wish to develop a better understanding of how boundary-spanning activity impacts consequential organizational dynamics. Given the strategic role boundary spanners play in activities designed to address important environmental interface issues, the quality of their interactions is an important element defining organizational effectiveness. As new forms of organization emerge, ever greater numbers of front-line personnel will be required to interact across organizational boundaries with enough authority to make important and timely organizational decisions (Drucker 1988; Peters 1988). Thus incentive exists for developing a better understanding of their work behavior and the factors which influence it. Future research built on the findings generated by this study could contribute to this process.

APPENDIX

MANAGERIAL BEHAVIOR QUESTIONNAIRE ITEMS

(rated on a 7-point scale from "Almost Never" to "Almost Always")

Promoting Information Flow

- My supervisor clearly communicates to subordinates the elements of the job that need to be accomplished.
- My supervisor communicates necessary information to people through a variety of channels.

Functioning Strategically

- Because of short-term pressures, my supervisor makes decisions designed to achieve quick results.

(Reverse scored)

- My supervisor discusses with subordinates the underlying causes of problems.
- My supervisor plans for the skills and knowledge required to meet future objectives.
- My supervisor makes decisions according to a well thought-out strategy.
- My supervisor talks about how organizational factors (e.g. policies, procedures, work styles) affect people in the company.
- My supervisor accepts short-term losses in order to achieve longer-term gains.

Developing Others

- My supervisor helps subordinates identify their needs, interests, and aspirations.
- My supervisor gives people accurate information regarding their performance.
- My supervisor helps subordinates identify their talents and skills.
- My supervisor provides training and/or personal growth experiences for the people who work here.
- My supervisor talks with subordinates to help them learn from their successes and failures.

APPENDIX (continued)

- My supervisor gives subordinates the "big picture" so they can understand how their work fits into a larger context.
- My supervisor teaches people what they need to know to do their jobs effectively.

Providing Guidance

- My supervisor talks with subordinates about the organization's goals, values, and standards.
- My supervisor rewards people for behaving in ways that reflect organizational values.
- My supervisor does subordinates' work for them. (*Reverse scored*)
- My supervisor asks for input on decisions from people who have expertise relevant to those decisions.
- My supervisor specifies the way a delegated task should be accomplished. (*Reverse scored*)

JOB DESIGN QUESTIONNAIRE ITEMS

(rated on a 7-point scale from "Very Inaccurate" to "Very Accurate")

Autonomy

- The job denies me any chance to use my personal initiative or judgment in carrying out the work. (*Reverse scored*)
- The job gives me considerable opportunity for independence and freedom in how I do the work.

Skill Variety

- The job requires me to use a number of complex or high-level skills.
- The job is quite simple and repetitive. (*Reverse scored*)

Task Identity

- The job is arranged so that I do *not* have the chance to do an entire piece of work from beginning to end. (*Reverse scored*)
- The job provides me the chance to completely finish the pieces of work I begin.

Task Significance

- This job is one where a lot of other people can be affected by how well the work gets done.
- The job itself is NOT very significant or important in the broader scheme of things. (*Reverse scored*)

Feedback from the Job

- Just doing the work required by the job provides many chances for me to figure out how well I am doing.
- The job itself provides very few clues about whether or not I am performing well. (*Reverse scored*)

GOAL EFFECTIVENESS QUESTIONNAIRE ITEMS

(rated on a 7-point scale from "Not at All" to "Totally")

Behavioral Specificity

- Do your work goals indicate what you should and should not be doing on the job?
- Do your work goals focus on the important aspects of your job?
- Do your work goals specify particular types of behavior for you to engage in on the job?
- Do your work goals have an impact on how you perform on the job?

Outcome Specificity

- How familiar are you with your work goals?
- Is it possible for you to determine the extent to which you have achieved your work goals?
- Do your work goals identify specific performance objectives for you to achieve?

Motivational Quality

- Is it a challenge to achieve your work goals?
- Do you find it personally rewarding to achieve your work goals?
- Are your work goals inspiring, in that they motivate you to work hard to achieve them?
- Is it important for you to try to achieve your work goals?

APPENDIX (continued)

Congruency

- Are your work goals consistent with other policies and practices in your organization?
- Do you believe that achieving the goals for you as an individual will help your work group achieve its goals?
- Do you believe that achieving your work group's goals will help the organization to achieve its goals?

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